

Rahul Raj

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Professional Summary

Data/Systems Engineer with 2+ years of industry experience designing data pipelines, automating training, validation, and test dataset workflows, and deploying machine learning systems. Proficient in Python, Pandas, SQL, and Azure cloud infrastructure, with hands-on experience in data labeling, dataset management, time-series signal processing. Currently pursuing an MSc in Artificial Intelligence at Dublin City University, deepening expertise in statistical data analysis, and ML pipelines and mobile data applications.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, R, MATLAB

Data Engineering: ETL Pipeline Design, Data Lake Architecture, Data Ingestion & Transformation, Data Labeling, Dataset Validation & Management, Workflow Automation, Time-Series Processing

Python Ecosystem: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, OpenCV, NLTK, FastAPI

Databases: PostgreSQL, MySQL, MongoDB (NoSQL), Redis; exposure to Azure Cosmos DB

Cloud & DevOps: Microsoft Azure (Event Grid, External Identities, DevOps), Azure APIs, AWS (S3/EC2), Git, GitHub Actions, Docker, Linux (scripting & automation), Windows

AI & Signal Processing: Machine Learning, DSP / Audio Signal Processing, NLP, Generative AI, LLM Integration, RAG Pipelines, Probability & Statistics

Platforms & Tools: Android (mobile data collection), ServiceNow, Genesys, Power BI, Postman, VS Code

Soft Skills: Cross-functional Collaboration, Technical Documentation, Agile Delivery, Analytical Thinking

Experience

Tata Consultancy Services – Systems Engineer

Jul 2023 – Aug 2025

Led data engineering and automation initiatives across two enterprise platforms, spanning ETL workflows, cloud data infrastructure, and AI-powered automation:

- Designed and maintained end-to-end ETL data synchronisation pipelines using cron jobs and Azure web job triggers, automating ingestion and transformation of knowledge base articles, chat logs, and call records across ServiceNow and downstream platforms – eliminating manual sync workload entirely.
- Built data labeling and summarisation scripts in Python to process and classify enterprise-scale document sets (sustainability reports) using TF-IDF and scikit-learn, improving ML classification accuracy across structured and unstructured data sources.
- Engineered RESTful APIs in Express.js and Node.js for MySQL & PostgreSQL data retrieval, applying SQL query optimisation to reduce frontend data loading time by 25%.
- Developed interactive data dashboards and visualisation pipelines using Angular and Node.js, reducing manual reporting effort by 80% and enabling real-time cross-platform data monitoring.
- Implemented Azure Event Grid triggers for real-time notification and data delivery pipelines at enterprise scale; managed secure data access via RBAC and token-based authentication, reducing processing overhead by 20%.
- Integrated Microsoft Azure External Identities with multi-platform OAuth (Google, LinkedIn) and SSO across enterprise applications.
- Deployed ServiceNow Virtual Agent using JavaScript and plugin automation, cutting repetitive ticket volume by 25%; automated an AI-powered service desk using Angular, Node.js, and Genesys APIs, achieving 50% reduction in average response time.
- Designed scalable microservices applying SOLID principles, OOP, and MVC architecture in Node.js & TypeScript across Linux-based environments, improving long-term code maintainability.

Tata Consultancy Services – Software Engineering Intern

Feb 2023 – May 2023

- Built an automated ESG data labeling and scoring pipeline in Python using NLTK, TF-IDF, and scikit-learn to classify and rank companies from sustainability reports; developed text preprocessing, feature extraction, and model validation workflows across enterprise-scale document sets.

Projects

ASR Training Dataset Pipeline – Low-Resource Telugu Speech Recognition

Python, PyTorch, Wav2Vec2 XLS-R, HuggingFace, CTC, Audio Signal Processing, Linux

- Designed and executed a full training/validation/test dataset lifecycle: ingested the MSCIL Telugu corpus (~44K audio samples), applied automated filtering (duration 1–15s), Telugu Unicode normalisation, and quality validation to produce a

clean 35K+ sample training dataset with documented splits, metadata, and reproducible preprocessing scripts.

- Built data labeling and annotation workflows including custom character-level CTC vocabulary construction, dynamic label masking, and token alignment validation – directly analogous to sensor data labeling and ground-truth dataset creation for algorithm development.
- Applied time-series signal processing techniques over raw audio waveforms: frequency-domain feature extraction, variable-length sequence handling, and noise-robust preprocessing using Wav2Vec2's convolutional encoder stack.
- Trained model on Linux (RTX A6000) with mixed-precision FP16, gradient checkpointing, and cosine LR scheduling; monitored WER/CER across training epochs and optimised toward sub-40% WER.
- Documented dataset schema, preprocessing decisions, and validation criteria to support reproducible algorithm development mirroring production-grade dataset management and documentation practices.

AI-Powered Image Verification System – Multi-Signal Data Pipeline

Python, FastAPI, Pandas, DINOv2, CLIP, SigLIP, XGBoost, Isotonic Regression, OpenCV, DCT/DSP

- Architected an end-to-end ETL data pipeline for image forensics: ingesting raw e-commerce image data, transforming it through a multi-signal feature extraction layer (semantic embeddings, DCT frequency-domain artefacts, PRNU noise residuals, edge-map structural features), and delivering calibrated predictions via FastAPI – achieving 93%+ F1 score.
- Designed and validated training, validation, and test dataset splits with rigorous quality checks; automated labeling workflows and dataset documentation to support reproducible model development.
- Implemented Isotonic Regression for probabilistic model calibration and built data summarisation scripts providing per-signal confidence scoring and interpretable decision outputs.

VectorSight – SVG Spatial Data Processing Pipeline | github.com/raahulraj1406/VectorSight

Python, FastAPI, Shapely, SciPy, LangChain, Next.js, Anthropic API, SSE Streaming

- Engineered a 14-step agentic data transformation pipeline converting raw SVG inputs into structured spatial representations via geometry processing (Shapely, SciPy), delivered in real time using SSE streaming via FastAPI.
- Built data extraction and transformation scripts to compute bounding boxes, centroid coordinates, containment hierarchies, and proximity relationships – converting unstructured visual data into rich, structured, LLM-consumable formats, mirroring sensor data transformation workflows.

Bhakti Saurabha – Automated Multilingual Data Processing Pipeline | rkmireland.org

Python, Claude API, Next.js 14, TypeScript, Tailwind CSS, Vercel, Structured JSON

- Engineered an automated data ingestion and transformation pipeline in Python to extract, translate, and structure 96 sacred hymns into 4 languages using the Claude API; designed a structured JSON data schema for metadata, language variants, and phonetic transliterations enabling fast client-side querying.
- Delivered a full-stack deployment on Vercel with CI/CD via GitHub; platform reached visitors from 11 countries within days of launch.

Education

Dublin City University, MSc in Artificial Intelligence

Sep 2025

Relevant Coursework: Machine Learning, Statistical Data Analysis, Data Analytics, Data Mining, Data Management & Visualisation, Computational Science, Artificial Intelligence.

AMC Engineering College, BE in Information Science and Engineering

Aug 2019 – Jun 2023

Capstone: Smart Classroom System integrating Python and Machine Learning for real-time automation and sensor-driven analytics in education.

Certifications & Training

Microsoft Azure AI Fundamentals (AI-900) | Anthropic MCP Certification (Model Context Protocol)

TCS Young Professionals Certification | Accenture Virtual Internship Program

Activities & Achievements

Marathon Finisher: Tata Mumbai Marathon (42.2K), Tata Ultra Marathon (50K, Lonavala), Procama Slam India Series (2024–2025), RunAddicts Marathon Series (2020–2025) – demonstrating endurance, discipline, and long-range goal commitment.